

Algorithm 3: Sorting Customer Orders

Exercise 3: Sorting Customer Orders

Scenario:

You are tasked with sorting customer orders by their total price on an e-commerce platform. This helps in prioritizing high-value orders.

Steps:

Understand Sorting Algorithms:

Explain different sorting algorithms (Bubble Sort, Insertion Sort, Quick Sort, Merge Sort).

Setup:

Create a class Order with attributes like orderId, customerName, and totalPrice.

Implementation:

Implement Bubble Sort to sort orders by totalPrice.

Implement Quick Sort to sort orders by totalPrice.

Analysis:

Compare the performance (time complexity) of Bubble Sort and Quick Sort.

Discuss why Quick Sort is generally preferred over Bubble Sort.